

Homework 0

Handed Out: Tue August 25

Due: Tue September 8, 8:00 p.m. ET

- The goal of written HW0 is to give you an idea of the mathematical background expected in this course. You should have seen all this material before; if you are rusty on any of it, this homework should encourage you to revisit it. HW0 does not comprehensively test *all* the prerequisites; instead it samples from areas that are relevant to this class to remind you to **brush up on those areas broadly**.
- Your performance in this class will depend significantly on your ability to understand mathematical descriptions of ML algorithms in the lectures. **If you find that you are not familiar with more than 30% of the material in this problem set, you should strongly reconsider taking the course.** Speak to the instructor / TAs if you have concerns. If you've seen this material before but just need a refresher, the course's own prereq primers cover exactly the material this problem set samples from:
 - **Linear algebra** (Questions 5, 6) · **Singular value decomposition**
 - **Probability** (Questions 2, 3, 4)
 - **Calculus** (Question 1) · **Vector geometry**

The full list, including a **Python, numpy and pandas primer** for the coding half and the **datasets** those notebooks use, is on the **course resources page**.

- Familiarize yourself with the “Human and AI Assistance in Homework” Policy included in the administrivia slides from the first lecture. <https://www.engineering.upenn.edu/~cis5190/fall2026/schedule.html>
- You are encouraged to format your solutions using L^AT_EX. You may make a copy of the template here and start adding your solutions: <https://www.overleaf.com/read/vjmrrbzbpmjv#f26f48>. This is a read-only link — make your own *private* copy before editing. Handwritten & scanned solutions are permitted, but remember that **if your work is deemed not legible by your grader, they will not give you points for it**, and we will not accept any post hoc explanations. You will submit your solution manuscript for written HW0 as a single PDF file.
- We will be using Gradescope to collect your homework assignments. Submit your solution manuscript as a PDF file via Gradescope. Your solutions won't be graded if you don't **clearly annotate which page each question is answered on, on Gradescope**.
- **If you are on the waiting list** to join the class, we recommend that you still work on and submit the assignment.

- **Grading.** The written HW0 and the coding HW0 are graded **separately**, and the same rule applies to each: as long as you answer more than **50%** of the written HW0 correctly you receive **full credit** for the written HW0, and likewise, more than **50%** on the coding HW0 earns **full credit** for the coding HW0. The per-question point values below (and the per-test-case scores the coding autograder reports) tell you how you did; they are not separately weighted in your final grade.

1 Background

These questions are graded **for being answered**, not for what you answer — there is no background that scores better than another, and none of this affects how your solutions are graded. We use the responses to calibrate the course and to know what to support. Students who joined from the waiting list answered these same questions on the waitlist form; we ask everyone so that we have a consistent picture of the class.

1. (1pt) [Linear algebra] Have you taken a university-level course covering matrix products, dot and cross products, rank, inverses, pseudoinverses, and eigenvectors? If so, which course?
2. (1pt) [Probability] Have you taken a university-level course covering random variables, joint and conditional distributions, priors and posteriors, expectations, and covariance matrices? If so, which course?
3. (1pt) [Multivariable calculus] Have you taken a university-level course covering partial derivatives, including matrix-vector derivatives such as $\frac{\partial(\mathbf{Ax})}{\partial \mathbf{x}}$ and $\frac{\partial(\mathbf{Ax})}{\partial \mathbf{A}}$? If so, which course?
4. (1pt) [Prior ML coursework] Have you taken a machine learning class before? If yes, which one, and how do you expect it to differ from what you anticipate in CIS 4190/5190?
5. (1pt) [Programming experience] Have you worked on coding projects with at least 100 lines of code? (Yes / No)
6. (1pt) [Python] Have you used Python before? (Yes / No)

2 Use of AI Tools

Report any use of AI tools (ChatGPT, Claude, Copilot, Gemini, Cursor, etc.) while working on **this written homework**. You report AI use for the coding homework separately, in the reporting cell at the end of the notebook.

Being honest here will not cost you points. This section is graded for being answered, not for any particular answer: reporting that you used no AI earns full marks, and so does reporting that you used a lot of it. We are trying to learn what actually helps people learn, which only works if the reports are accurate.

1. (2pts) Did you use AI tools on this written homework? If **no**, say so and you are done with this section. If **yes**, also answer:
 - Which tools did you use?
 - **How** did you use them? Be specific: checking an answer you had already derived, explaining a concept you were stuck on, generating a first draft you then rewrote, doing the derivation for you, etc.

- How do you think this affected your learning of this material? Say so if you think it helped, and equally if you think it let you skip understanding something you would otherwise have worked out.

3 Time Spent

1. (2pts) Roughly how long did each part of this **written** homework take, in hours? Decimals are fine ($0.5 = 30$ minutes), and enter 0 for any question you skipped. Report a time for each question as well as for reading the assignment. Rough estimates are genuinely useful — please do not agonize over precision, but do fill this in **as you go** rather than reconstructing it at the end. Also graded for being answered, not for what you answer.

4 Multiple Choice & Written Questions

Note: You do not need to show work for multiple choice questions, but questions that say “show your work” are graded on the reasoning, not just the final answer. If formatting your answer in L^AT_EX, use the solutions template linked at the top of this document.

1. [Calculus] Let $f(x, y) = 3x^2 + y^2 - xy - 11x$
 - a. (4pts) Find $\frac{\partial f}{\partial x}$, the partial derivative of f with respect to x . Find $\frac{\partial f}{\partial y}$. Select one of the below four options.
 - A) $\frac{\partial f}{\partial x} = 6x - 6y - 10, \frac{\partial f}{\partial y} = y - x$
 - B) $\frac{\partial f}{\partial x} = x - y - 11, \frac{\partial f}{\partial y} = 2y - x$
 - C) $\frac{\partial f}{\partial x} = 6x - y - 11, \frac{\partial f}{\partial y} = 2y - x$
 - D) $\frac{\partial f}{\partial x} = x - 6y - 11, \frac{\partial f}{\partial y} = 2y - x$
 - b. (7pts) Now consider a *different* function, $g(x, y) = e^x + x + y^2$. Find **all** optima of g over $\mathbb{R}^2$: state how many there are and what kind each one is (maximum, minimum, or saddle point), giving any coordinates to two decimal places. Show your work.
 Format your answer as: [Number]; [Type: (x, y)]; [Type: (x, y)];
 ...
2. [Probability] Assume that the probability of obtaining heads when tossing a coin is λ .
 - a. (4pts) What is the probability of obtaining the first head on the $(k + 1)$ -th toss? Select one of the below four options.
 - A) $(1 - \lambda)^k \lambda^k$
 - B) $(1 - \lambda) \lambda$
 - C) $(1 - \lambda) \lambda^k$
 - D) $(1 - \lambda)^k \lambda$

- b. (4pts) What is the expected number of heads if the coin is tossed k times? Select one of the below four options.

A) $k \times \lambda$
B) k^λ
C) λ^k
D) $(1 - \lambda)^k$

3. [Probability] Assume we know the following facts about COVID-19 testing:

- 95% of infected people test positive. This is called the true positive rate.
- 0.01% of uninfected people test positive. This is the false positive rate.
- Based on population-level statistics in Pennsylvania, we expect that among all people who are tested, 10% are infected.

With this information, please answer the two questions below:

- a. (4pts) John, who lives in Pennsylvania, tests positive. What is the probability that he is infected?

A) ≈ 0.999
B) ≈ 0.099
C) ≈ 0.5
D) 1

- b. (4pts) With all other values remaining the same, what should the false positive rate change to, for John's probability of being infected to change to 0.95?

A) $\approx 0.556\%$
B) $\approx 0.056\%$
C) $\approx 0.1\%$
D) $\approx 1\%$

4. [Probability] The variance of a random variable X is defined as

$$\text{Var}(X) = E[(X - E[X])^2] .$$

- a. (6pts) Let X and Y be random variables with $E[X] = 5$, $E[Y] = 2$, $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, and correlation $\rho_{XY} = 0.5$. Compute $E[XY]$, and state the formula you used.

Format your answer as: [Number]; [Brief formula]

- b. (9pts) Prove that $\text{Var}(X) = E[X^2] - E[X]^2$. Please show your work in detail.

5. [Linear Algebra] Let

$$\mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$
$$\mathbf{M} = \begin{bmatrix} 4 & 7 \\ 5 & 8 \\ 6 & 9 \end{bmatrix}$$

- a. (4pts) What is the length of the vector $\mathbf{v}$, $\|\mathbf{v}\|_2$ (sometimes written as $\|\mathbf{v}\|$)? Select one of the below four options.
- A) $\sqrt{3}$
 - B) -1
 - C) $\sqrt{5}$
 - D) 5
- b. (4pts) What is the result of the multiplication $\mathbf{v}^T \mathbf{M}$, where T indicates the transpose operation? Select one of the below four options.
- A) Undefined, sizes do not align.
 - B) $\begin{bmatrix} 0 & 0 \\ 5 & 8 \\ -12 & -18 \end{bmatrix}$
 - C) $\begin{bmatrix} -3 & -6 \end{bmatrix}$
 - D) $\begin{bmatrix} -7 & -10 \end{bmatrix}$
- c. (4pts) What is the result of the multiplication $\mathbf{M} \mathbf{v}^T$? Select one of the below four options.
- A) Undefined, sizes do not align.
 - B) $\begin{bmatrix} 0 & 0 \\ 5 & 8 \\ -12 & -18 \end{bmatrix}$
 - C) $\begin{bmatrix} -3 & -6 \end{bmatrix}$
 - D) $\begin{bmatrix} -7 & -10 \end{bmatrix}$

6. [Linear Algebra] Let

$$\mathbf{A} = \begin{bmatrix} 4 & 2 \\ 2 & 7 \end{bmatrix}$$

- a. (10pts) Compute the eigenvalues **and** eigenvectors of $\mathbf{A}$ by hand. Please show your work in detail.
- Format your answer as: Eigenvalues: [low, high]; Eigenvectors: [v1], [v2]
- b. (4pts) Find the maximum value of $\mathbf{x}^T \mathbf{A} \mathbf{x}$ for any unit vector $\mathbf{x}$.

5 Python Programming Questions

A Google Colab notebook is linked in the “HW0 Coding” assignment on the class website. This will tell you everything you need to do, and provide starter code. See the recording from the first day of class for how to use Colab and how to submit homeworks.

The coding HW0 is worth **65 points** in total, and is graded separately from the written HW0 (see the grading note above):

- **50 points** are graded automatically by the PennGrader cells inside the notebook. Running a grading cell submits that answer and records your score immediately. **Your most recent submission for each grading cell is the one that counts** — re-running a cell replaces your previous score for that test case, even if the new score is lower.
- **15 points** are for the `matplotlib` figure in Question 4 of the notebook, which is graded by hand from the copy you embed in *this* written solution PDF (see below). Do not forget to embed it — there is no way to award these points otherwise.
- Submit the notebook as `.ipynb` file to the coding assignment on Gradescope.
- Remember to download and embed the figure for question 4 into the *written problem set solution pdf*.